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SUPERSEDING
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FEDERAL SPECIFICATION

DEICING-DEFROSTER FLUID (AEROSOL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers an aerosol deicing-defroster fluid to aid in the removal of frost and thin ice from vehicle windows (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-E-489 - Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces).
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

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Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substances Act Regulations.
49 CFR 170-179 - Department of Transportation, Hazardous Materials Regulations Board.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards:

D 56 - Flash Point by Tag Closed Tester.
D 1123 - Water in Engine Coolant Concentrate by the Iodine Reagent Method.
D 1177 - Freezing Point of Aqueous Engine Antifreeze Solution.
D 3069 - Delivery Rate of Aerosol Products.
D 3094 - Seepage Rate of Aerosol Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Chemical Specialities Manufacturers Association (CSMA):

Aerosol Guide

(Application for copies should be addressed to the Chemical Specialities Manufacturers Association, 1001 Connecticut Avenue, NW., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Finished product. The finished product shall consist of a deicing-defroster fluid packaged with a suitable propellant in an aerosol dispenser.

3.2 Net weight of dispensable contents. When tested as specified in 4.3.1, the aerosol dispenser (see 3.5) shall contain not less than 14 ounces avoirdupois (397 grams) of dispensable contents at the time of first acceptance by the Government.

3.3 Deicing-defroster fluid. The deicing-defroster fluid shall meet the physical and chemical requirements of table I, when tested as specified in 4.2.4. The deicing-defroster fluid shall be miscible in water.

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TABLE I. Physical and chemical requirements

Property	Requirement	
	Minimum	Maximum
Water (percent by weight)	--	48.0
Flash point (Tag Closed Cup)	26°C	--
Dissolved solids (percent by weight)	--	0.1
Freezing point (25 percent solution)	--	-8.0°C

3.4 Prohibited material. The finished product shall not contain vinyl chloride monomer.

3.5 Aerosol dispenser. The aerosol dispenser shall consist of a can, a valve assembly, and a cover to prevent accidental discharge of the contents.

3.5.1 Can. The can shall conform to PPP-C-96, type IX, class 2. The can shall not be corroded by the contents.

3.5.2 Valve assembly. The valve stem shall be fitted with a side discharge plastic actuator. The direction of discharge shall be indicated either by the shape of the actuator or by a mark on it. The valve assembly shall produce a spray when activated by finger pressure and automatically shut off when the pressure is removed. The valve assembly shall be designed and located so that when activated, none of the spray shall impinge on any part of the dispenser. When the valve is actuated, the contents shall be ejected only through the discharge orifice.

3.5.3 Cover. The dispenser shall be protected from accidental discharge by a rigid protective device. The protective device may be a snug-fitting cap of metal or rigid plastic, or it may be a rigid structure designed to prevent accidental discharge of the contents.

3.5.4 Directions for use. The dispenser shall be marked or labeled with the manufacturer's normal directions for use.

3.6 Delivery rate. The delivery rate of the full dispenser at 70°F (21°C) shall be not greater than 8 grams per second (g.p.s.) when tested as specified in 4.3.3.

3.7 Leakage. When tested as specified in 4.3.2, no more than 1.0 milliliter (ml) of gas shall leak from the filled dispenser.

3.8 Discharge form. At a temperature of 12°F (-11°C), the contents shall be dispensed in the form of a spray.

3.9 Spotting on paint. When tested as specified in 4.3.10, the deicing-defroster fluid shall cause no more than slight spotting on a painted surface.

3.10 Smearing of windshield. When tested as specified in 4.3.5, the deicing-defroster fluid shall not cause smearing of the windshield.

3.11 Dispenser stability at 130°F (54°C). The filled dispensers shall not exhibit leakage, distortion, or other pressure-related defects or damage after being heated at 130°F (54°C) for a period of 1 hour (see 4.3.8).

3.12 Laws and regulations. The finished product and its marking and labeling shall comply with Federal Hazardous Substances Act Regulations and Department of Transportation, Hazardous Materials Regulations Board regulations.

3.13 Shelf life. During storage for a period of 1 year under general warehouse conditions at temperatures between 0°C and 44°C, the aerosol dispensers shall not be corroded by the contents.

3.14 Warranty. The contractor shall provide a warranty that the product meets the requirement of 3.13.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Inspection of the end item. Components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, or qualified in this specification or applicable purchase documents.

4.2.2 Examination of preparation for delivery. Packaging, packing, and marking shall be examined to determine compliance with the requirements of section 5. Defects shall be scored in accordance with table II. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-2, Acceptable Quality Level (AQL) of 4.0 expressed in terms of percent defective. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be all shipping containers of deicer-defroster fluid presented for delivery at one time.

TABLE II. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper, or missing.

4.2.2.1 Examination of closure, waterproofing, and banding of containers. When shipping containers are required to be closed, waterproofed, and banded in accordance with PPP-B-636, examination for defects in closure, waterproofing, and banding shall be in accordance with the appendix to PPP-B-636.

4.2.3 Testing of the filled dispensers. The filled dispensers shall be tested as specified in table III. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-4, AQL of 2.5 percent defective. The sample unit shall be one filled dispenser. The lot shall consist of all dispensers filled in the same batch and presented for delivery at one time.

TABLE III. Instructions for testing the filled dispensers

Property Tested	Specification Reference		Results Reported	
	Requirement	Test Method	Pass or Fail	Numerically to Nearest
Leakage	3.7	4.3.2		0.1 ml
Delivery rate	3.6	4.3.3		0.1 g
Spray impingement on container	3.5.2	4.3.3	X	
Discharge other than through discharge orifice	3.5.2	4.3.3	X	
Discharge form	3.8	4.3.4	X	
Smearing of windshield	3.10	4.3.5	X	
Net weight of dispensable contents	3.2	4.3.1		0.5 g

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4.2.4 Testing of the deicer-defroster fluid. The deicer-defroster fluid shall be tested as specified in table IV. The lot shall consist of all filled dispensers which have been filled with deicer-defroster fluid which has been manufactured in the same batch and which dispensers are presented for delivery at one time. Four dispensers shall be selected at random from the lot, the propellant released, and the deicer-defroster fluid combined and mixed to form a composite sample. The composite sample shall be stored in a closed container. Failure of the composite sample to pass any of the tests in table IV shall be cause for rejection of the lot.

TABLE IV. Instructions for testing the deicer-defroster fluid

Property Tested	Specification Reference		Results Reported
	Requirement	Test Method	
Water content	table I	ASTM D 1123	to nearest 0.1 percent
Flash point	table I	ASTM D 56	to nearest 0.5 degree C
Dissolved solids	table I	4.3.6	to nearest 0.1 percent
Freezing point	table I	4.3.7	to nearest 0.1 degree C
Spotting of paint	3.9	4.3.10	as pass or fail

4.2.5 Testing of the filled dispensers for stability at 130°F (54°C). The filled dispensers shall be tested for stability at 130°F (54°C). The lot shall consist of all dispensers presented for delivery at one time. The sample unit shall be one filled dispenser. The number of samples specified below shall be selected randomly from the lot. Test the dispensers in accordance with 4.3.8. Failure of any of the dispensers to pass the requirement of 3.11 shall result in rejection of the lot.

Lot size	Sample size
2 - 15	2
16 - 50	3
51 - 150	5
151 - 500	8
501 - 3,200	13
3,201 - 35,000	20
35,001 - 500,000	32
500,001 up	50

4.3 Test methods.

4.3.1 Net weight of dispensable contents. Weigh each dispenser to the nearest 0.1 gram. Perform the tests specified in 4.3.2, 4.3.3, 4.3.4, and 4.3.5. Then keeping the lower half of the dispenser submerged in ice water and shaking the dispenser occasionally, spray the contents of the dispenser into a sink or other appropriate container until the dispenser is completely discharged. Reweigh the dispenser to the nearest 0.1 g. Record the difference in weights as the net weight of dispensable contents.

4.3.2 Leakage. Determine the leakage in accordance with ASTM D 3094, Seepage Rate of Aerosol Products.

4.3.3 Delivery rate. Determine the delivery rate in accordance with ASTM D 3069, Delivery Rate of Aerosol Products. Report the average of three determinations. During the performance of this test, observe if any part of the spray impinges on the dispenser or if any of the contents are discharged other than through the discharge orifice.

4.3.4 Discharge form. Cool the dispenser to 12° ± 2°F (-11° ± 1°C). Press the actuator. The contents shall be discharged in a spray.

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4.3.5 Smearing of windshield. Thoroughly clean the windshield of an automobile with a spray-on, wipe-off cleaner. Put on windshield wiper blades which have not been used except for this test. Pour a cup of water over one-half of the windshield and immediately spray the same half of the windshield thoroughly with the deicing-defroster fluid and turn the windshield wipers on at the slow speed. The windshield shall not be smeared with the deicing-defroster fluid. Rinse deicing-defroster fluid from automobile finish.

4.3.6 Dissolved solids. Place approximately 5 g of deicing-defroster fluid in a tared weighing bottle. Weigh to the nearest 0.1 mg. Place the bottle and contents in a circulating air oven and heat slowly to 105°C. If fluid remains in bottle, heat slowly to 210°C. Dry to constant weight. Record residue as dissolved solids.

4.3.7 Freezing point. Prepare a 25 percent solution by volume of the deicing-defroster fluid in distilled water. The deicing-defroster fluid shall be miscible in the distilled water. Measure the freezing point of this 25 percent solution in accordance with ASTM D 1177 - Freezing Point of Aqueous Engine Antifreeze Solution. ASTM D 1177 may be modified as follows: 1) acetone may be used as the bath liquid; and 2) the solution may be placed in a beaker and stirred with a magnetic stirrer and stirring bar.

4.3.8 Dispenser stability at 54°C. Submerge dispensers in an automatic control, circulating, constant temperature bath at room temperature. Place a safety shield in front of the bath in case a dispenser ruptures. Program the bath for 54°C and allow the temperature to rise. Keep the dispensers in the bath at 54° ± 1°C for a period of 1 hour after the bath temperature has reached 54°C. Using leather gloves and a face shield, remove the dispensers from the bath and examine them. There shall be no leakage, distortion, or other pressure-related defects.

4.3.9 Optional test for vinyl chloride. This test may be used to verify compliance with 3.4. The manufacturer is not required to perform the test. Insert the aerosol can in a modified can piercing pressure measuring device (CSMA Aerosol Guide: "Method for Internal Pressure Determination of Aerosol Products") which has a 1/8 inch Swagelock fitting. Pierce the can in a closed hood since a can may occasionally tear when pierced. Connect the apparatus and can to a gas sampling valve on a programmable gas chromatograph. Operate the gas chromatograph under the following conditions: gas sampling valve temperature - 110°C; helium carrier flow - ca 80 ml/min; programmed temperature rise of 6°C per minute from 100°C to 200°C with a 2-minute post injection hold at 100°C; flame ionization detector temperature - 240°C. Use 1/8 inch outer diameter stainless steel columns packed with 60-200 mesh silica gel.

Use a known sample of vinyl chloride to calibrate the gas chromatograph. Run the sample through the gas chromatograph. There shall be no detectable material in the sample having the same retention time as vinyl chloride.

4.3.10 Spotting of paint. A test panel with a layer of enamel conforming to TT-E-489 shall be prepared and air dried or baked and air dried as specified in paragraph 4.3.21 of TT-E-489. Place a minimum of 10 drops of deicing-defroster fluid on the panel. Allow to stand undisturbed for 16 hours at 4° ± 1°C. At the end of that time, rub the spots with a soft cloth dampened with distilled water. If spots remain, rub them with a soft cloth and a typical automotive cleaner. After this treatment, spots which can be described as "slight spotting, visible only in certain lighting" are permitted. If spots are easily visible, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Twelve (12) filled dispensers specified in 3.2 and 3.5 shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed, waterproofed, and banded in accordance with the appendix to the box specifications.

5.1.2 Level B. Twelve (12) filled dispensers specified in 3.2 and 3.5 shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

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5.1.3 Level C. Filled dispensers, in quantities as specified (see 6.2), shall be packed in a container complying with the Uniform Freight or National Motor Freight Classification, as applicable. The container shall be packed to assure carrier acceptance and safe arrival at destination.

5.2 Marking. The dispensers and shipping containers shall be marked in accordance with 3.5.4 and Fed. Std. No. 123 for civil agencies, or MIL-STD-129 for military agencies, as applicable.

6. NOTES

6.1 Intended use. The deicing-defroster fluid covered by this specification is intended to be sprayed on vehicle windows to aid in the removal of frost and thin coverings of ice.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.1).
(c) Quantities required (see 5.1).
(d) Applicable marking document (see 5.2).

MILITARY INTEREST

Review Activity:

DSA-GS

CIVIL AGENCY COORDINATING ACTIVITIES

CPSC
GSA-FSS

Preparing Activity: GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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